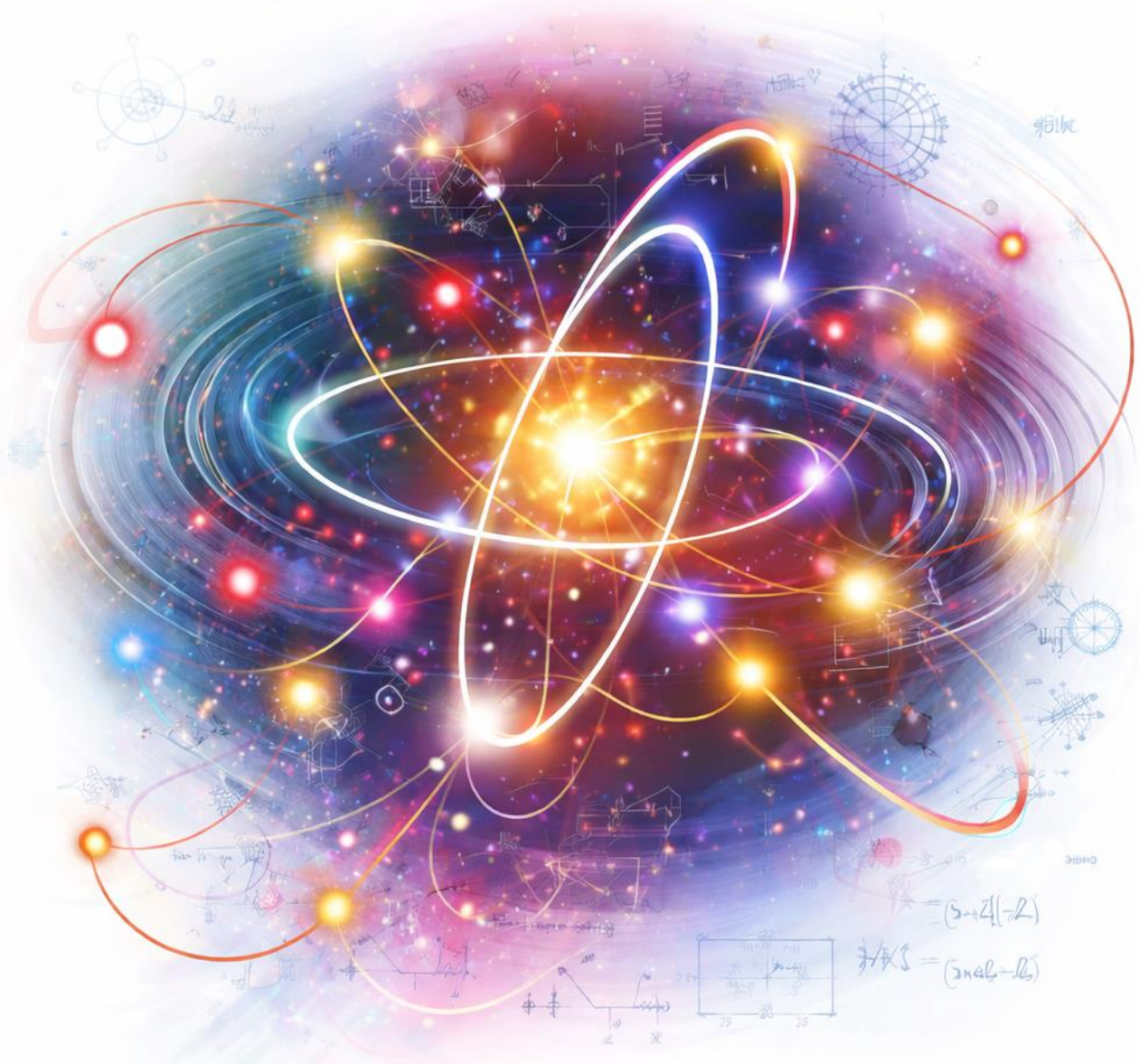


QED

Quantum ElectroDynamics



NEWSLETTER
VOLUME 1, 2025



No.4: 2020-2025 Overview for QED

This issue marks the final update of Volume 1 of the QED newsletter before transitioning to Volume 2. Volume 1 spans 22 pages and documents the progress of the QED project over the past five years.

Our newsletters are designed to be concise, accessible updates—suitable for online distribution and printed handouts at events—highlighting recent developments, investment planning, and foundational project information.

If you are interested in learning more about the QED project, please contact us via email, phone, or through the “Contact Us” page on our website (QED-NE.com). We also have a [Patreon](#) in which you can donate to us monthly to keep us going!



Figure 1: Model 24A with new connectors.

Variable Electromagnetic Drives Advancing Toward Space Applications

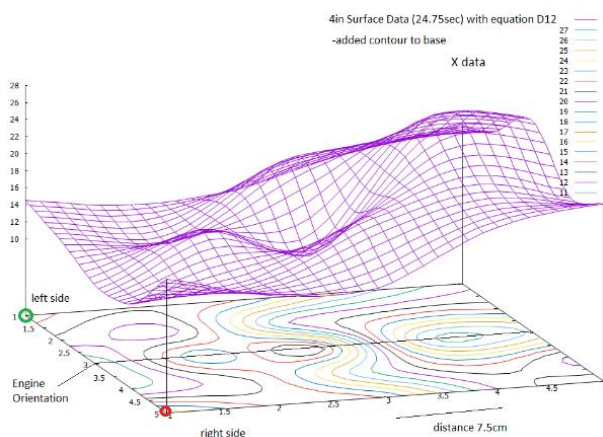


Figure 2: Detailed mapping of the gravitational disturbances at 10 watts.

watts of electrical power—a performance milestone that significantly exceeds conventional electric propulsion efficiency.

Quantum Electro Dynamics (QED) is actively developing **Variable Electromagnetic (VEM) Drive technology** designed for operation in both terrestrial and space environments. The VEM Drive generates a directional pulling force through controlled electromagnetic interactions that compress spacetime in front of the drive, producing measurable thrust without the use of propellant.

High-resolution mapping of the VEM Drive’s warp signature has revealed critical insights into how the system behaves under laboratory and simulated space conditions. Most notably, VEM Drive Model 17 has demonstrated a pulling force of 3.37 pounds (15 newtons) in any orientation while consuming only 1,500



Foundational Technology Briefing

The VEM Drive was invented by Professor David Pares, whose work explores the use of electromagnetic energy to induce controlled spacetime compression. To date, no other propulsion system has demonstrated artificial spacetime warping using electromagnetic energy alone.

Over 24 generations of VEM Drive prototypes have been constructed and tested by the QED team. Each model has demonstrated exponential improvements in pulling force while simultaneously reducing the energy required per unit of lift.

Thermal imaging conducted during active operation revealed an unexpected and notable result: the VEM Drive exhibited a 4.5°F cooling effect relative to ambient air temperature while engaged—suggesting non-classical energy interactions worthy of further investigation.

Advances in QED VEM Technology

QED currently has three VEM Drive architectures in active prototyping and development:

- **Model M13-O**
A solid aluminum, 12-inch grid design optimized for CubeSat and small satellite missions in Low Earth Orbit (LEO).
- **Model M24-A**
A two-foot copper fractal grid designed to handle higher input power levels, enabling substantially increased thrust capability. This model is currently under active development.
- **Model M25-A**
A three-dimensional, sixth-harmonic fractal grid architecture engineered to dramatically improve the lift-to-power ratio, increasing pulling force while reducing overall energy demand.

Each successive model represents a step change in efficiency, scalability, and system performance.

Independent Verification and Academic Evaluation



Figure 3: Reid Sherman

In a major milestone for QED, a NASA Fellowship Senior Engineering student, Reid Sherman, will conduct independent verification testing of VEM Drive Models 13 and 24 during the Fall and Spring academic semesters. All testing will take place at the University of Nebraska–Lincoln Mechanical Engineering Laboratory, with final documentation and analysis expected by May 2026.

This marks the first time QED has opened its core propulsion system to independent third-party evaluation, reflecting confidence in the repeatability, rigor, and transparency of our experimental results.



Experimental Validation

1. Interferometer Proof-of-Concept Experiments

These experiments proved that there was something happening within the tripole field that was created in the lab. 140 sets of images were collected. Read our AIAA paper for more information. It can be found on our website. Pares, D. P., & Finley, K. (2018)

Once these results were found, it showed the QED team that there was something happening with our device/created fields that needed to be refined to increase performance, which is why we decided to do the next experiments.

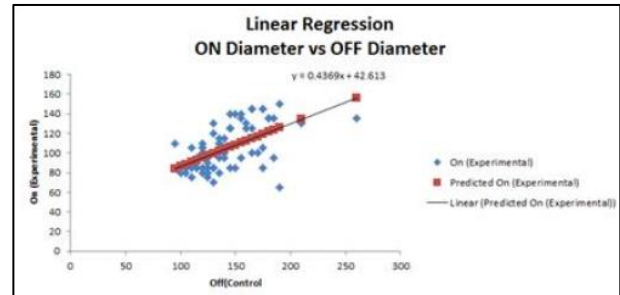


Figure 4: Interferometer linear regression data plot.

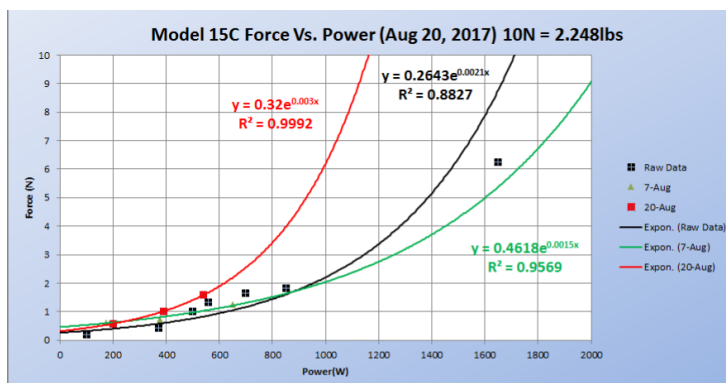


Figure 5: Shows our previous 2017 projections to which the newer drives meet or exceed the predictive models.

2. Force Measurement and Iteration

These experiments were done with ferrous and non-ferrous materials over several years. The most recent experiments have been posted to YouTube and analyzed. This experiment resulted in 11.7N at 1500W of power with Model 19, the 2ft aluminum solid state drive. Our previous experiment was 4.6N at 1500W of with Model 17, which is a 2ft drive made from copper. We have also tested with our smaller old Model 13 which produced .88N at 500W.

Multiple designs have been used throughout the past 15 years. This shows us that as long as the key elements are incorporated into the drive, the device will perform the force creating action. What the QED team has been working on is finding the most optimal design patterns to boost the performance for our small scale testing with our limited funding. If we had more funding we could be testing all different power levels and designs, but since this is mostly self funded we have had to be very careful with what we spend money on and what hardware we can use to experiment with.



2025 Space Engine Comparisons

Space Engines	 Moon	 Mars	 Pluto	 Proxima Centauri
Nasa	3 Days	260 days	12.5 years	133,027 years
Space X Starship	2 Day	90 days	6.47 years	46,109 years
Nuclear Propulsion	1 Day	45 days	3.23 years	22,016 years
VEM Drive	15 minutes	6 days	146 days	2,700 years
Speed of Light	1.2 seconds	12 minutes	1 day	4.24 years

Figure 6: This shows engine travel times to common destinations. The time shown for the VEM Drive is just with what has currently been produced in the lab. Theoretically, the VEM drive may be able to travel at the speed of light or faster based on relativity.

Advantages of the VEM Drive over ION and Hall Effect Thrusters

1. Fully powered by DC Energy

Can be completely powered by batteries, solar, nuclear, or other electrical generators. Not true of other engines.

2. Propellantless

Does not need to be refueled like all other engines

3. Higher Theoretical Speeds

Can reach speeds far beyond other engines

4. Compact and Scalable Design

Can be used in small satellite designs or larger vehicles

5. Heat-Free, Solid-State Design

The VEM is solid-state, thus increasing its lifespan



QED Phase I Monetary Potential

Phase I: Earnings/Savings Differences on Satellites					
Satellite Type	Weight Range	Example Cost	Increased Performance Earnings (+50%)	Extended Life Savings (+100%)	Multiplied Profits (+300%)
Cube Sat (1U-16U)	1-32kg	\$50,000	\$25,000	\$50,000	\$150,000
Small Sat	1–500 kg	\$300,000	\$150,000	\$300,000	\$900,000
Large Sat	2,000–6,000+ kg	\$100 million	\$100 million	\$100 million	\$300 million
High Complexity Sat (Hubble & James Webb Space Telescope)	Varies	\$1 billion+	\$500 million	\$1 billion+	\$3 billion+
Constellation Sat (Star-link, One Web)	Varies (SmallSats)	\$1 million	\$500,000	\$1 million	\$3 million

Table 1: Phase I earnings table that looks at savings/earning for satellite-based operations.

This table highlights the financial advantages of equipping satellites with small VEM Drives operating under 500 watts—technology **already demonstrated** in lab conditions and now targeting a space-ready MVP. The primary financial gains come from two key areas: enhanced orbital positioning (which boosts performance by 50%) and extended satellite lifespan (doubling operational years), both enabled by the precise, fuel-free maneuverability of VEM propulsion.

These benefits translate directly into increased profitability across satellite types. CubeSats, typically costing around \$50,000, could generate an additional \$150,000 in value. SmallSats could see earnings rise to \$900,000 per satellite. For large satellites, the value jumps to \$300 million, and for high-complexity satellites like Hubble or James Webb, profit potential reaches over \$3 billion. Constellation satellites (e.g., Starlink) also benefit significantly, with potential multiplied earnings of \$3 million per unit.

These are conservative estimates, as a VEM-driven satellite with an enduring energy source could theoretically remain in optimal orbit indefinitely—limited only by the lifespan of its sensors or communication hardware. This makes VEM technology not only a cost-saving measure but a fundamental upgrade to the entire lifecycle economics of space infrastructure.

Additionally, incorporating VEM Drives into satellite platforms enables entirely new mission profiles. Satellites can shift orbits on demand, avoid collisions autonomously, and maintain constellations with minimal energy input. This agility turns satellites from passive observers into active, maneuverable assets—enhancing mission resilience, reducing replacement needs, and enabling dynamic, real-time space infrastructure management.



QED Phase II Monetary Potential

Phase II: Savings on a \$60M Space X Launch			
Cost Type	Percent of total	Dollar Amount	Phase II savings
Launch Vehicle Production and Ops	65%	\$39M <i>80% is propulsion system and tanks</i>	\$27.3M
Propellant	3%	\$1.8M	\$1.8M
Ground Operations	15%	\$9M	
Mission Assurance and Insurance	7.5%	\$4.5M	\$450,000
Payload Integration	7.5%	\$4.5M	
Regulatory and Compliance Costs	2%	\$1.2M	
1 Launch	100%	\$60M	\$29.5M per launch
30 Launches		\$1.8B	\$885M per 30

Table 2: The Phase II cost breakdown of based on a \$60 million SpaceX launch reveals substantial savings through the implementation of Variable Electromagnetic (VEM) propulsion.

Here we would save mostly on the Launch Vehicle Production costs, Propellant, and Mission Assurance and Insurance costs. This is because our VEM drives are easy to create, easy to control, and they use no fuel. This contrast is the main reason this technology will be so useful in the future when it comes to space transportation.

Traditional systems allocate 65% of launch costs to vehicle production—\$27.3 million of which is propulsion-related. VEM drives eliminate much of this expense due to their lightweight, fuel-free design. They also remove the 3% cost allocated to propellant (\$1.8 million), since they require no fuel. Additionally, mission assurance and insurance—currently 7.5% of costs—can be reduced thanks to the safety and reliability of electromagnetic systems. When scaled to 30 launches, VEM systems could save up to \$885 million, reducing total launch costs nearly in half. These savings highlight the transformative potential of VEM propulsion for space logistics.



Future Phases – 10 Year Plan

QED Three Phases

Phase I – \$5 million (3 years)

- VEM Production Lab
- Satellite Testing of VEM Drive
- MVP will be complete for (small revenue)



Phase II – \$50 million (5 years)

- Complete a functional Earth to Space VEM Drive
- Can then sell services for lifting cargo to space (big revenue)
- Will have government approval / restricted airspace

Phase III - \$500 million (10 years)

- Complete a Spaceport on Earth for VEM Drives
- Completed Large scale craft that can take people
- Long distance space missions, beyond the Solar System



Forecasted Private LP Share Structure & Secondary Market Strategy

Overview

This page outlines a structured approach for selling private Limited Partnership (LP) shares through phased valuations, projected growth, and secondary market exit options. The model is designed to balance early-stage risk with long-term upside while preserving company control.

Limited Partnership Structure

Total Shares: 100

Total Shares	Allocation
30 Shares	Held by the company (long-term control and governance)
10 Shares	Flexible allocation (strategic partners, future needs)
60 Shares	Investor shares across three funding phases

Investor Phases

Phase	Shares	Capital Raised	Price per Share	Post-Money Valuation
Phase 1 – Pre-Seed	20	\$5 million	\$250,000	\$25 million
Phase 2 – Seed	20	\$50 million	\$2.5 million	\$250 million
Phase 3 – Series A	20	\$500 million	\$25 million	\$2.5 billion

Fractional Ownership

To improve accessibility and liquidity:

- **1/5 share:** \$50,000
- **1/10 share:** \$25,000

Projected Growth by Phase (Table 2)

The model targets a **1000 percent total return per phase**, scaled to each stage's time horizon.

Example: 1/10 Share Investment

Phase	Timeline	Avg. Annual Return	Initial Investment	Estimated Value
Phase 1 – Pre-Seed	3 years	~333 percent per year	\$25,000	\$250,000
Phase 2 – Seed	5 years	~200 percent per year	\$250,000	\$2.5 million
Phase 3 – Series A	10 years	~100 percent per year	\$2.5 million	\$25 million

Liquidity Timing

Secondary market sales may occur after each completed funding phase, subject to partnership terms and buyer availability.

Capital Use

Capital is allocated toward technology development, validation milestones, talent acquisition, and strategic partnerships to support valuation growth across phases.



No.3: Summer Update 2024

This year, we have constructed four Model 13 configurations. The second M13 we tested a new solderless connector configuration. A new power grid was designed for the third M13 with wider power feeds and expanded the vertical fractal layout. The fourth M13 used a copper paste to make the connections solderless to the VEM Drive. This provided a very stable tuner with no power reflections and allowed numerous tests without any adjustments to the tuner.

The fifth Model 13 is being finished for testing this week. We have experimented with making a nano-diamond copper past formula and this has reduced the resistivity even more in the VEM Drive. The fifth M13 will use this formula and new power connections and mountings for the connectors to the engine.

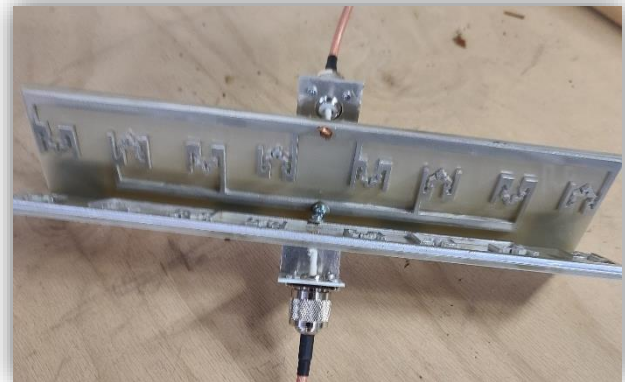


Figure 1: Model 13C with new connectors.



Figure 2: M13 configuration vertical

In summary, advances have been made in equipment and VEM drive stability. We have tested experiments between 100 to 500 watts of power consistently with no power reflection. This is a very important advancement for remote drive operations. We have also experimented with single and dual feed connectors to the drives. Dual connectors work the best for the 12-inch motor configuration. The original M13 can still obtain a 90 gram pull at 500 watts or .86N, which was achieved this June. The original is thirteen inches long, so the new grids had to be modified and were reduced from the model 19 configuration (which produced 11.73 N and was 2 feet in length) to fit on a 12-inch format. This next set of experiments should see if we can produce the expected 1N at a 500-watt power level.



Figure 3: Left to Right, model 13, 13B, 13C.

Update on Direction

In 2023 we were able to get two very good tests completed and recorded. These were the June 9, 2023 and the December 15, 2023 tests that we posted on YouTube. While those were our most dramatic results with 4.7N and 11.7N at 1500W, we had a total of 41 other tests logged. They had an average of 32.5g. This summer we have been working on getting an older model, Model 13, to be more consistent than what we have achieved in the past. If this is done, we plan to work with a team of satellite engineers to add our VEM Drive to a CubeSat and test it in space. That is why we have been working on so many new M13 designs.

The goal this year is to produce 35g as consistent as possible with 500W of power.

The plan for summer/fall 2024:

1. Build a standardized drive of Model 13, which is about 1ft across, 300mm.
2. Optimize the system to increase the percentage of successful runs above 63%.
3. Create 35g of force with 500W with the new system / new drive.
4. Work to secure funding for going to space on a CubeSat 2025.

Chart 1: *Fall 2023 Summary (June 13 – December 15).*

Successful Tests	Total Tests	Success Percentage
26	41	63%
	Average grams displaced =	32.5g

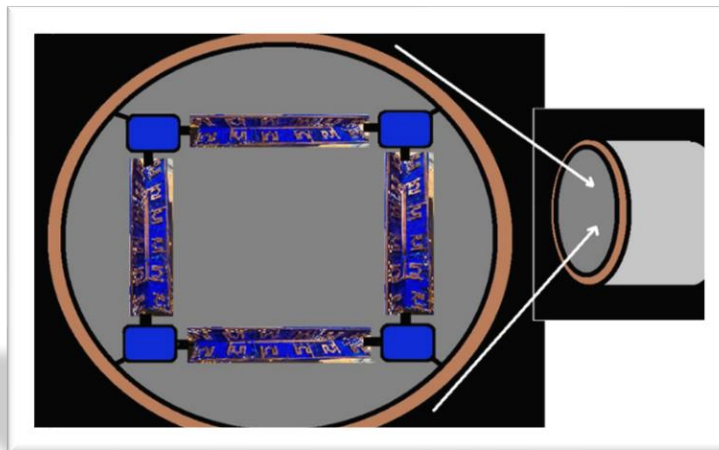


Figure 4: Proposed Quad Drive

Update on Quad Design

Since our last newsletter we have been able to put together a rig that will hold 2 to 4 Model 21A drives together in a square rig as proposed. We tried some limited testing with 2 of the drives in a smaller fashion, but due to limitations in space the drive will not work in our small faraday cage. Therefore, testing of this configuration will have to wait until we have moved the testing to a lab or outside the lab.

Progress in 2024

This year we have built 4 new models with 2 more to be done soon. One will be the finalized version for satellite testing and the other will be a larger version with pairs of fractals together as seen in figure 3. All testing will be done by September as we will await future funding. The winter newsletter will have all the updates and more details on the test result and direction of the project. So far this year we have done 50+ complete experiments, which is more than we have ever been able to do in a year in the past. These, as well as last years experiments, are thanks to being partially funded by private donations.

Patreon Started

At the start of the year we also started a Patreon going with no content, mainly for a few people to donate if they wish. We are looking into uploading some content by the end of the year such as our old/new ideas and some old concept art. Stay tuned!

Thanks to our 6 supporters and 2 followers out there!

https://www.patreon.com/user?u=111856331&utm_medium=clipboard_copy&utm_source=copyLink&utm_campaign=creatorshare_fan&utm_content=join_link

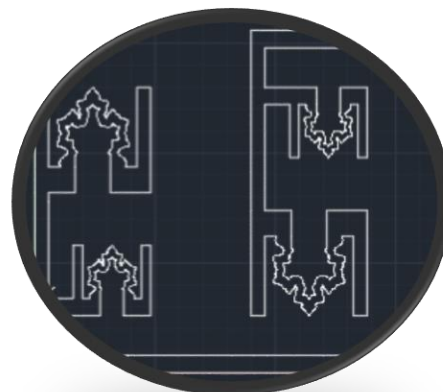


Figure 5: Proposed double configuration to be tested before the end of the year.



Space Warp Related Papers with Summary

1. "Covariant quantum field theory of tachyons."

By Paczos, Jerzy, et al. Physical Review D 110.1
(2024): 015006.

A recent paper suggests that tachyons, hypothetical particles that move faster than light, might actually exist, challenging the prevailing belief among physicists that such particles are impossible. Traditionally, faster-than-light particles are thought to violate causality by potentially traveling back in time or possessing negative energy. However, the new paper proposes a novel mathematical approach that could reconcile these issues by interpreting particles as having both forward and backward time components, thus avoiding negative energy problems.

This perspective also raises possibilities for phenomena like wormholes, warp-drives, and anti-gravity, which are currently deemed improbable due to their association with negative energy. The speaker highlights that this development could offer new insights into the Fermi paradox and the potential for advanced extraterrestrial civilizations.

<https://arxiv.org/pdf/2308.00450>

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2. "What no one has seen before: gravitational waveforms from warp drive collapse."

By Clough, Katy, Tim Dietrich, and Sebastian Khan.
arXiv preprint arXiv:2406.02466 (2024).

We are on the cusp of a new era in gravitational wave astronomy, with detectors like LIGO having already identified numerous black hole and neutron star mergers. The next generation of detectors, including Cosmic Explorer, the Einstein Telescope, and the Laser Interferometer Space Antenna (LISA), promises to expand our observations further. A recent study speculates that these advanced instruments might even detect signs of extraterrestrial technology. Specifically, it explores whether gravitational waves could be generated by the collapse of a warp bubble, a hypothetical construct from science fiction that bends spacetime to allow faster-than-light travel.

While practical warp drives may face immense challenges, including exotic matter requirements and massive energy needs, the study suggests that a collapsing warp bubble could produce detectable gravitational waves. Although current detectors like LIGO are not yet sensitive enough to catch such signals, future advancements could potentially reveal these extraordinary phenomena, providing insights into both advanced alien technologies and the limits of general relativity.

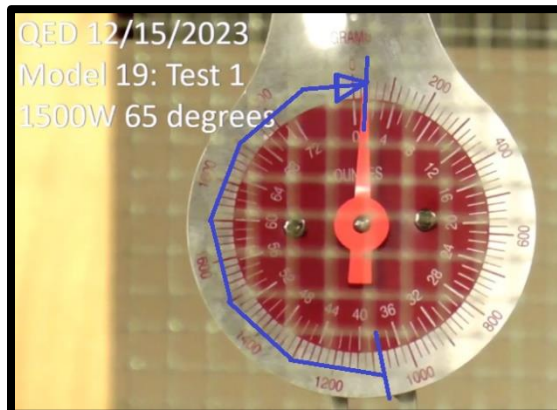
<https://arxiv.org/pdf/2406.02466>

If you know someone that we can contact for additional funding or publications send an email to David Pares at dppares@outlook.com and/or Kyle Finley at kfinley.ion@gmail.com.



No.2: Our Latest Experiment

December 15, 2023



This December we were able to do a test that blew our socks off! We were able to record a pull on a scale of 1196 grams. It was such a great force that the scale broke halfway through the test. This test was with Model 19, with a total force of **11.73 Newtons in 20 seconds**.

Here is a clip of the experiment. This has been loaded to YouTube for review.

<https://www.youtube.com/watch?v=kNJBIGiDLO>

This is the second main pulling experiment that we recorded and posted online this year (2023). In June we were using a 10N scale that had 10g per tick. Since we maxed out that one, this time we decided to try the 20N scale, with 20g per tick. In the past the 20N scale had too much resistance to move it with the force that we were producing. We were able to overcome that with this test and prove that model 19 can outproduce model 17.

There is more to model 19 than being more efficient. This model is our solid-state model with almost no solder joints. That means that the timely repair that we used to have to do on model 17 is gone. It also means that it is easier to produce as a single cutout. Lastly, it was made from aluminum, which is easier to cut than copper and also cheaper.

With all of these benefits we see this model as the main model to use in our next phase, which will be discussed on the next page. We plan to do some more testing with model 19 before the next phase of our research. We will be replicating our last test as well as doing a lift test and maybe testing different power levels with this drive. We also have some new equipment that will enhance our testing capabilities for the new year.



Figure 1: Model 19 VEM Drive.

QED would like to wish everyone a Merry QED Holiday Season!

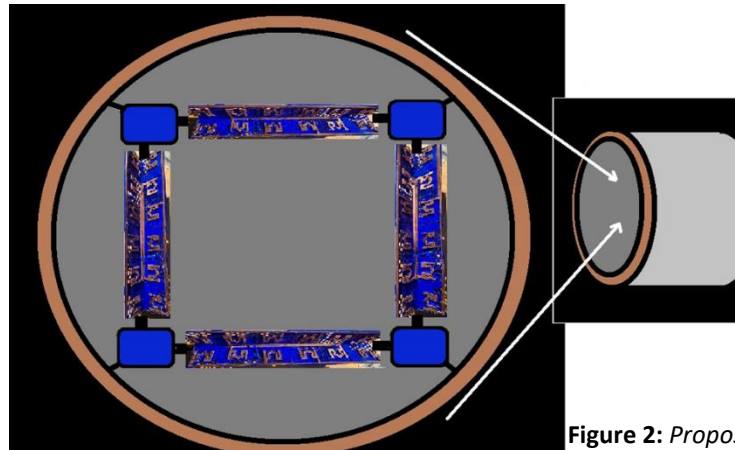
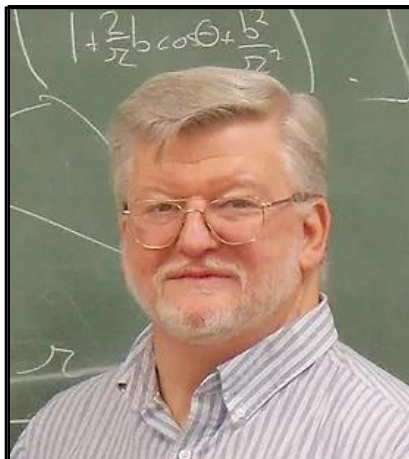


Figure 2: Proposed Quad Drive

Update and Future Steps

Testing of new drive designs has been very intense since June of this year. Over 52 testing cycles have been initiated and completed with very promising results. Our other promising design and innovation is the evaluation and testing of our new articulating drive harnesses, which allows up to four drives to be activated up all at once.

This is the first time that the clustering of engines has been possible. Each engine type has a known focal point in the compress of the fabric of space. This also varies somewhat with the amount of RF power input. Tests have revealed that the power in, to work out is exponential. With the articulating harness this will make for some exciting results, with crossing the fields of the engines in the harness. We are looking for a magnification of total work output in these upcoming testing cycles. Power for this test will be provided by our upgraded linear amps and custom final combiner, which will take us up to over 5000 watts of power.



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Chart 1: 151% improvement between June and December's test

	Grams (g)	Newtons (N)	Pounds (lbs)
Model 19	1196	11.728	2.635
Model 17	475	4.66	1.048
Difference	721	7.068	1.587

Our Future Steps

While the December test was excellent, we noticed that half ways through the test the VEM Drive maxed out the scale and slowed down (seen in figure 3). The velocity should increase to the end of the experiment and not drop off halfway through. We believe that by buying a new scale and running a full test we might be able to double the output or more. This would be more than enough to lift the drive, which would lead to an exciting lift test.

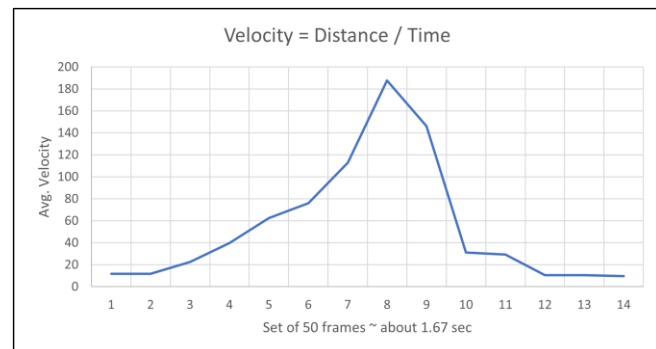


Figure 3: Graphed Velocity from December 16 Test

The next steps we plan to pursue:

1. Replicate and run a full test downwards.
2. Replicate the test in a lift setup.
3. See how different and higher power levels improve performance.
4. Create a quad lifter with 4 model 19s.

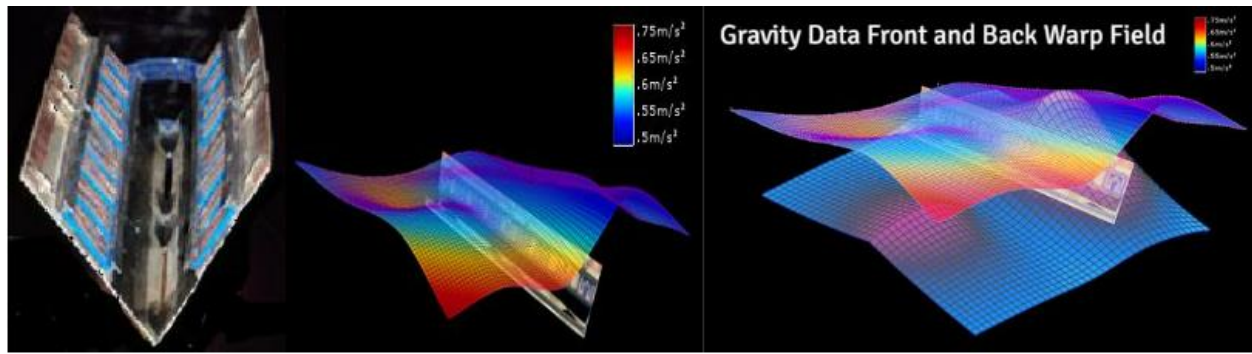
Our Future Needs

The next 3 steps can be performed in our current setup, but once we get to the full quad lifter model, we will be looking at getting a larger lab to contain the larger VEM Drive setup/batteries/wires and the full assembly. This would include a walk-in faraday cage and better setups for the cameras and monitoring stations for power/reflected power.

We also will have a Patreon for QED-NE to raise some additional funds and depending on how much support we get we may give subscribers monthly updates. But to start out we will be seeing how much we can raise from just donations as we are always focusing mainly on improving drive performance.

https://www.patreon.com/user?u=111856331&utm_medium=clipboard_copy&utm_source=copyLink&utm_campaign=creatorshare_fan&utm_content=join_link

If you know someone that we can contact for additional funding or publications send an email to David Pares at dppares@outlook.com and/or Kyle Finley at kfinley.ion@gmail.com.



No.1: Our Latest Experiment

June 09, 2023

We are thrilled to announce our latest experiments in the field of quantum electrodynamics! Our Variable Electro Magnetic (VEM) Drive has made significant progress and achieved a substantial lift of Model 17, with a total force of **4.66 Newtons** in 18 seconds.

Here is a clip of the real experiment. This has been loaded to YouTube for review.

<https://www.youtube.com/watch?v=Wl0oeyuBzg8>

This was the first time that we were able to see direct movement on to the mechanical scale during a live experiment. This is just one additional piece of evidence that our system is producing multiple Newtons of Force! In the future we plan to update our force vs power chart and build a quad drive.

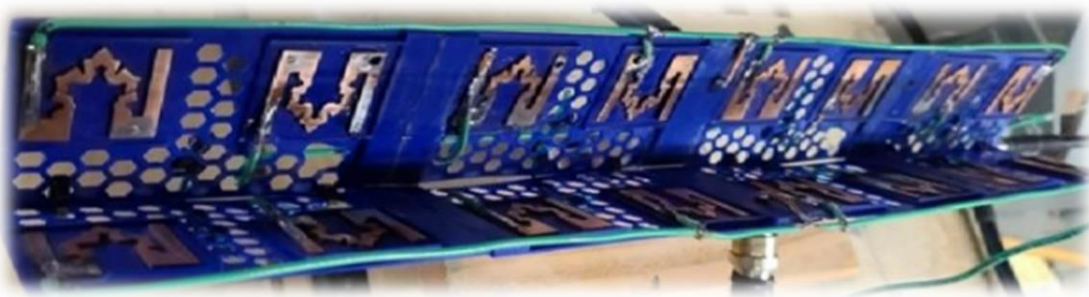
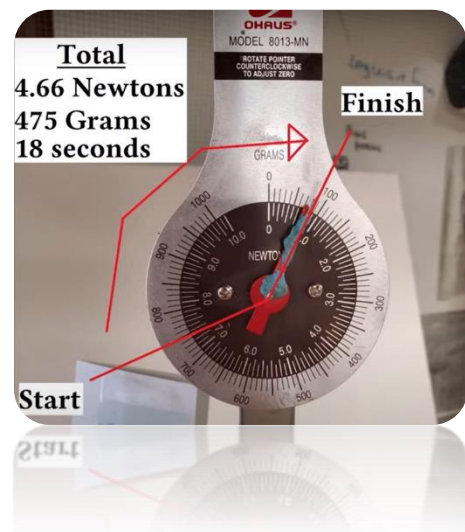
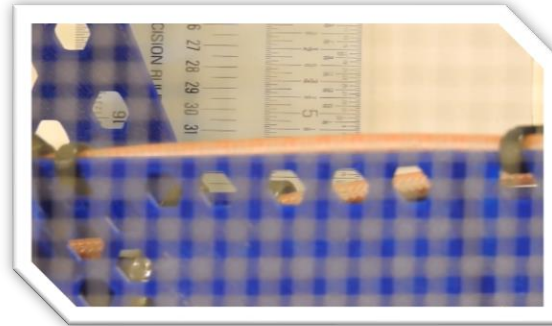


Figure 1: Model 17 VEM Drive.



Experiment #6

April 30, 2023 - 12:34 PM



We are thrilled to announce our latest experiment in the field of quantum electrodynamics! Our Variable Electro Magnetic (VEM) Drive has made significant progress and achieved a substantial lift of Model 20, with a total force of 6.7168 Newtons.

There is a 17 second raw clip of the actual experiment. This has been loaded to YouTube for review.

<https://www.youtube.com/watch?v=Kdax3im25AY>

This was significant progress in achieving a substantial lift of Model 20 at 4.7625 mm in 13 seconds.

Total force 6.7168 Newtons.

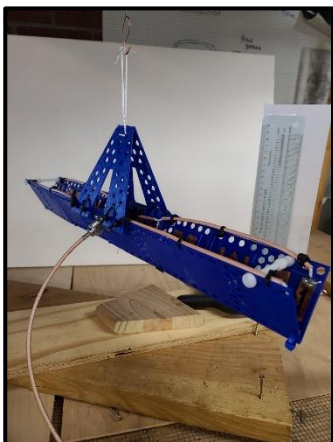


Figure 2: Model 20 VEM Drive.

Data Specifications:

- Total duration of experiment for lift is 13 seconds.
- Power 1500 Watts, 0 Watts of reflective power SWR 1.06
- Power cables can handle 7,000 Watts of power.
- Model 20 Weight .81 lbs. power cables .70 lbs. total 1.51 lbs.
- Ambient air temp 55 ° F
- Air around Model 20 55 ° F after test
- Lift 3/16 ", 4.7625 mm or 6.7168 N of force. 1500 watts 13 sec.

Velocity = .3663 mm/sec

Acceleration = .028 mm/sec²

Using the current Model 20 technology to go to space:

A 100 Kg spacecraft would take 25 days to travel to Mars, covering 48.676 million miles with a maximum speed of 72,000 meters per hour, or 162,000 miles per hour.

A 100 Kg spacecraft would take 206 days to travel to Pluto, covering 3.3 billion miles with a maximum speed of 600,000 meters per hour, or 1.386 million miles per hour. That is 0.0019 C or 0.19% speed of light.



Project Information (July 2023)

Background Information:

For the last ten years, the QED team has been developing a proprietary Variable Electromagnetic (VEM) Drive, invented by David Pares, that compresses the fabric of space. Our VEM Drive has direct application for terrestrial and space operations that avoids the use of fuel (including fossil fuels of any kind). We have applied for and have received a provisional U.S. Patent to protect this intellectual property and we are pursuing seed funding at this time for the next phase of our work.

The project is entirely self-funded at the QED laboratory in Omaha, Nebraska. We have performed replicable demonstrations of our VEM Drive at just 1,500 watts energy and can produce up to (15 N) 3.37 pounds of pull solely on battery power. The latest NASA ion drive uses 6,000 watts of power and only produces .243 newtons of thrust. Our VEM Drive currently could travel to Mars in 18 days. The next step is to construct a more powerful set of drives capable of achieving autonomous lift, to not only lift materials, but to do it inexpensively to access space.



Figure 3: Model 18 of the VEM Drive.

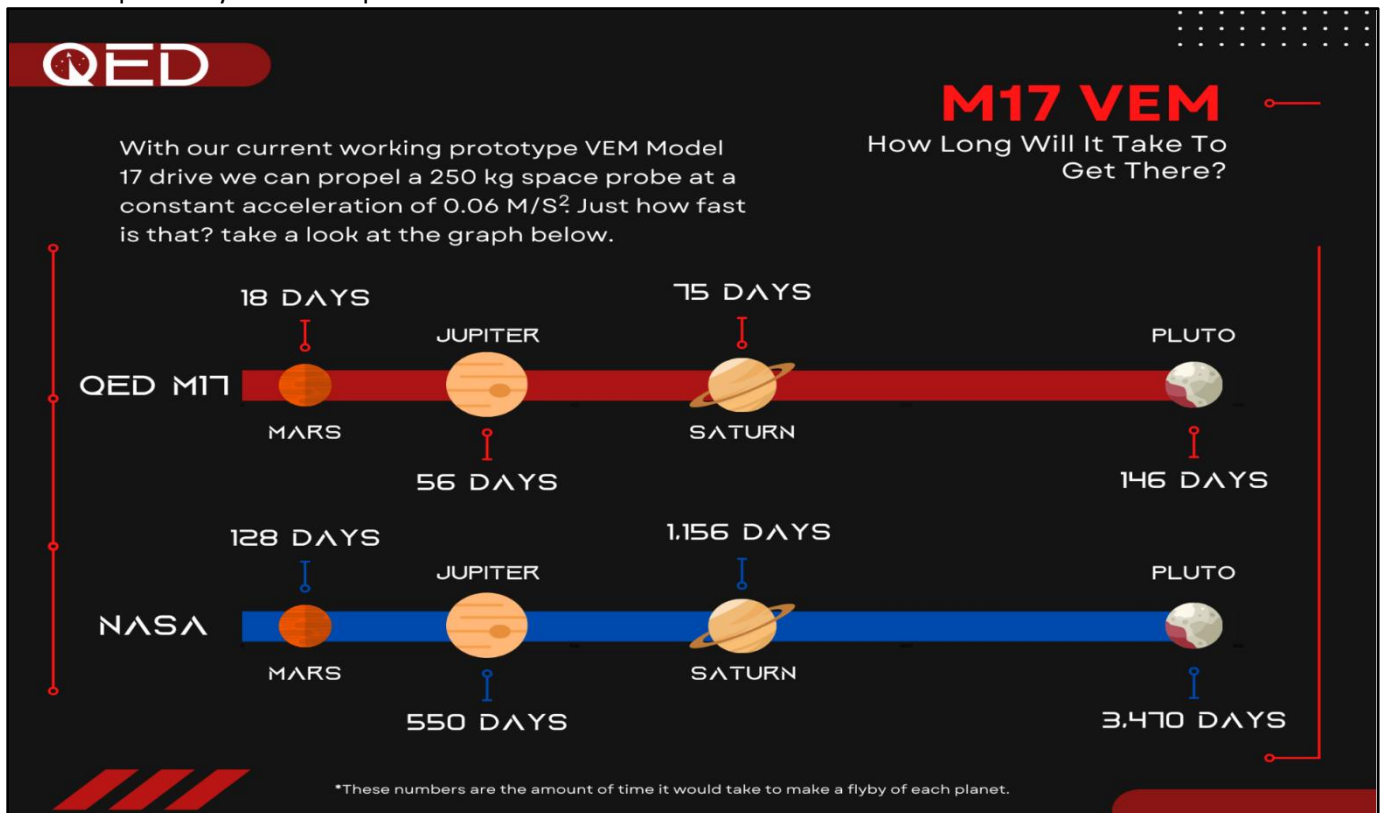


Figure 4: This image shows travel times to Pluto using our model 17's results vs Nasa's best.



QED's Strategic Goal

QED's strategic goal is to maintain America's economic growth in a space-based economy. The VEM Drive technology has uses for traveling in the solar system, station keeping, artificial gravity production in spacecraft, or space debris removal. The ability to utilize this technology in these applications will help maintain America's global standing in the space industry by lowering costs of transporting goods on Earth as well as in space. Lowering costs will help develop a space faring economy creating a great expansion in Earth and space economies. This technology is entirely electric which helps address climate change and reducing atmospheric pollution. We believe this technology is of great interest to the space community and to the preservation of the planet.

Warp signatures around VEM Drive

The Variable Electro Magnetic Drive (VEM) Drive creates a pulling action by compressing the fabric of space in front of the drive using electromagnetic energy. The mapping of the warp signature shows critical aspects of how the VEM Drive works in space and in a terrestrial environment. We are the first and only team that is creating an artificial warping of spacetime using electromagnetic energy. Many VEM Drive models have been built by the Quantum Electro Dynamics (QED) team, each one exponentially increasing its energy to lift ratio.

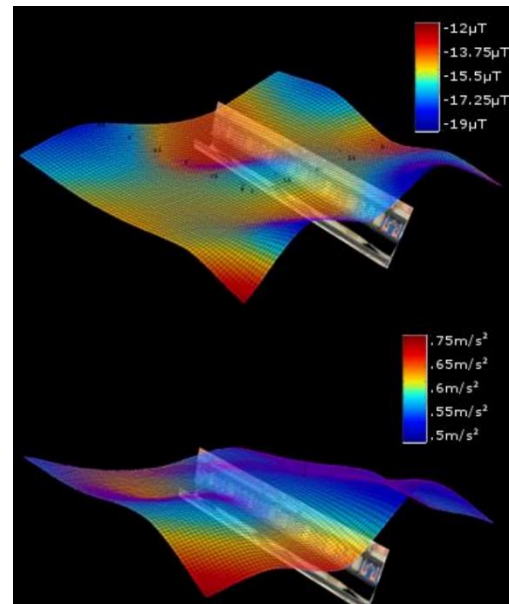
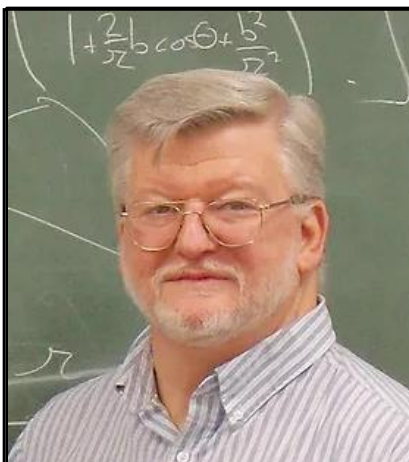


Figure 5: Shows 4-in Magnetic Flux (Top) and Gravitational data (Bottom).



Profile: David Pares

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Additional Charts and Graphs

Here we have our previous charts that show exponential results with additional power input, our interferometer experiment, and our historical timeline over the past 10 years.

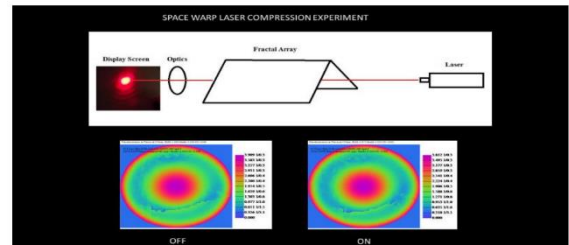


Figure 2. Space Warp Laser Compression experiment.

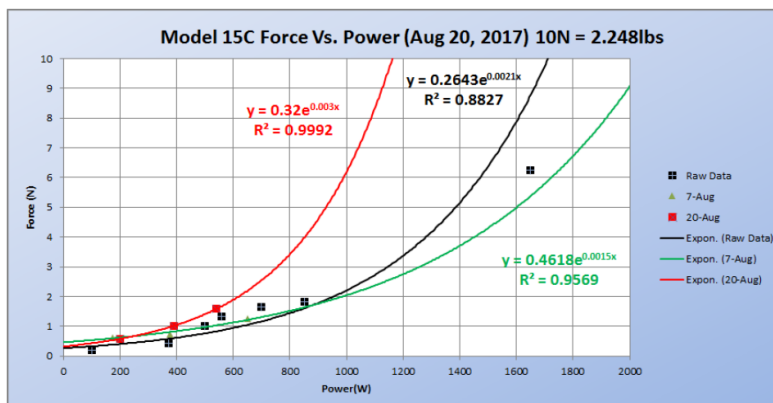


Figure 6: 2017 exponential force vs power graphs.

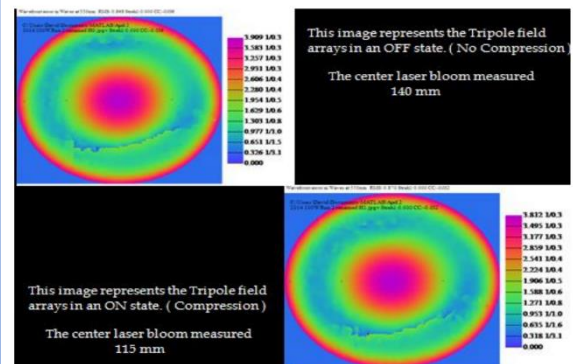


Figure 7: Interferometer results.





Comparison to other Space Engines

Technology	Newtons	Seconds	Watts
Dawn Nstar*	0.091	3100	2100
NASA NEXT ION	0.237	4170	6000
NASA EM**	0.012	24s - over 60s	80
Hall Thruster*	0.6	60	10000
Japan's Hayabusa 2 (ION)*	0.01	2800	1200
QED VEM Drive (2018)M16	6.26	20	1650
QED VEM Drive (2021)M17	15	20	1500

Figure 8: Chart comparing the difference in Newtons produced by different drives.

In this chart, it can be seen that our VEM drives greatly outperforms any of the other space engines currently being used. QED has continued to work on increasing the efficiency between the power used and force created over the last ten years. While we have not lifted our drives into space yet, in outer space these drives would be very effective in allowing spacecraft to traverse the solar system at their current capability. Once funded, these drives could be used to travel between Earth and space as well as for interstellar travel.

Our Future Steps

Our next step is to cluster four drives together to try and produce a more expansive and powerful field over all the drives. The hope is that with 4500W and four drives working at the same time we can create a very visible lifting effect that can be seen in 30 seconds of testing. We will also be creating new force vs power charts with the new equipment that we will be getting soon.

Our Future Needs

This project needs to be pushed forward at a faster pace to ready the US and the world for a space-based economy. While we are able to keep making progress at our current speed, there are hurdles such as getting the VEM drive space certified and tested that are currently out of reach for the company. There also needs to be manufacturing improvements to standardize each of the VEM Drives to perfection. Software development needs to be done so that the drive is not controlled manually with a throttle and electrical feedback can be monitored in real time. Lastly, a secure lab setting is needed to protect the drives and provide a manufacturing and development area to advance this project.

If you know someone that we can contact for additional funding or publications send an email to David Pares at dppares@outlook.com and/or Kyle Finley at kfinley.ion@gmail.com.



Executive Statement

As the Executive of our visionary organization, I am proud to share our exciting mission to revolutionize space travel through cutting-edge research and development in Quantum Electrodynamics (QED). Our organization is at the forefront of pushing the boundaries of science and technology to unlock the potential of space warp, artificial gravity, and other inventions as well.

At our organization, we believe that QED holds the key to unlocking the power of science and enabling breakthroughs in space travel that were once only in the realm of science fiction. Our team of pioneering scientists, engineers, and researchers are dedicated to advancing the frontiers of QED and harnessing its potential to revolutionize space travel.

One of our key areas of focus is space warp, a concept that involves bending the fabric of space-time to achieve faster-than-light travel. We believe that space warp has the potential to generate new possibilities for interstellar travel, allowing us to explore distant galaxies and unlock the possibilities of universal travel in ways never before imagined.

In addition, our organization is committed to developing artificial gravity technologies that could transform human space exploration. Artificial gravity has the potential to mitigate the detrimental effects of prolonged space travel on human health and enable long-duration space missions, making human exploration of the cosmos more feasible and sustainable.

Furthermore, our organization is dedicated to developing other innovative space technologies such as non-ferrous manufacturing, fusion, and wireless energy transfer.

QED is driven by a passion for pushing the limits of what is possible in space travel and pioneering a space-based economy. We are committed to conducting groundbreaking research, collaborating with leading experts in the field, and driving technological advancements that will shape the future of space, science, and exploration.

In conclusion, as the Executive of our visionary organization, I am honored to lead a team of brilliant minds in our pursuit of revolutionizing space travel through QED. We are dedicated to pushing the boundaries of science and technology to develop space warp, artificial gravity, and other technologies that will change the way we explore and travel through space. Together, we will continue to push the limits of human knowledge and create a brighter future for space exploration.



Kyle Finley

Chief Executive Officer
Quantum Electro Dynamics